§1 Hammond's Practice Midterm 2

1. A quotient of a polynomial space.

a) $E(p+q) = \big(p(0)+q(0),\, p(1)+q(1)\big) = E(p)+E(q)$ and $E(cp) = cE(p)$. Surjective: $E(1-x) = (1,0)$ and $E(x) = (0,1)$, so $E\big(a(1-x)+bx\big) = (a,b)$.

b) $U = \{p : p(0) = p(1) = 0\}$. Such p is divisible by $x(x-1)$, so $p = x(x-1)(c+dx)$. Basis: $x(x-1) = x^2 - x$ and $x^2(x-1) = x^3 - x^2$. Check: $\dim U = \dim \mathcal{P}_3(\mathbb{R}) - \dim \operatorname{range} E = 4 - 2 = 2$.

c) Well-defined: if $p + U = q + U$ then $p - q \in U = \operatorname{null} E$, so $E(p) = E(q)$. Linear: $\tilde{E}\big((p+U)+(q+U)\big) = \tilde{E}\big((p+q)+U\big) = E(p)+E(q)$, and similarly for scalars. Injective: $\tilde{E}(p+U) = 0$ means $p \in \operatorname{null} E = U$, so $p + U = 0 + U$. Surjective because E is. (This is the first isomorphism theorem.)

d) $(1-x)+U$ and $x+U$, with $\tilde{E}\big((1-x)+U\big) = (1,0)$ and $\tilde{E}(x+U) = (0,1)$. They form a basis because $\tilde{E}$ is an isomorphism and their images are a basis of $\mathbb{R}^2$. (Equally good: $1+U$, $x+U$ with images $(1,1)$, $(0,1)$.)

2. Dual maps.

a) $T^*\colon W^* \to V^*$, $T^*(\ell) = \ell \circ T$. It lands in V^* because a composition of linear maps is linear. Linear: $T^*(\ell_1+\ell_2) = (\ell_1+\ell_2)\circ T = \ell_1 \circ T + \ell_2 \circ T$, since $(\ell_1+\ell_2)(Tv) = \ell_1(Tv) + \ell_2(Tv)$; and $T^*(c\ell) = c(\ell \circ T)$.

b) For $\ell \in Z^*$: $(S\circ T)^*(\ell) = \ell \circ S \circ T = (S^*\ell)\circ T = T^*(S^*\ell) = (T^* \circ S^*)(\ell)$.

c) If $T^*\ell = 0$, then $\ell(Tv) = 0$ for every v. Since T is surjective, every $w \in W$ is some Tv, so $\ell(w) = 0$ for all w and $\ell = 0$. So $\operatorname{null} T^* = \{0\}$.

d) $(T^*\ell)(x,y) = \ell\big(T(x,y)\big) = 3(2x-y) = 6x - 3y$.

3. Upper-triangularization.

a) If $\dim V = 1$, take any basis v_1. Then $Tv_1 \in V = \operatorname{span}(v_1)$, so the 1×1 matrix is upper-triangular.

b) If $u = (T-\lambda I)w \in U$, then $Tu = T(T-\lambda I)w = (T-\lambda I)(Tw) \in U$, since T commutes with $T - \lambda I$. Since λ is an eigenvalue, $T - \lambda I$ is not injective; by rank–nullity it is not surjective, so $\dim U < \dim V$.

c) If $U = \{0\}$, then $T = \lambda I$ and every basis works. Otherwise $T|_U$ is an operator on the nonzero complex space U of smaller dimension, so by the induction hypothesis U has a basis $u_1, \ldots, u_m$ in which $\mathcal{M}(T|_U)$ is upper-triangular. By Prop. 59 that means $Tu_j = T|_U u_j \in \operatorname{span}(u_1, \ldots, u_j)$ for each j.

d) Extend to a basis $u_1, \ldots, u_m, v_1, \ldots, v_r$ of V. For each k, $Tv_k = (T-\lambda I)v_k + \lambda v_k$ with $(T-\lambda I)v_k \in U = \operatorname{span}(u_1, \ldots, u_m)$, so $Tv_k \in \operatorname{span}(u_1, \ldots, u_m, v_1, \ldots, v_k)$. Together with (c), every basis vector is sent into the span of itself and the earlier basis vectors, so by Prop. 59 the matrix of T in this basis is upper-triangular.

4. A consequence of the minimal polynomial.

a) Write $m_T = x^d + a_{d-1}x^{d-1} + \cdots + a_1 x + a_0$ with $a_0 = m_T(0) \neq 0$. From $m_T(T) = 0$:

$$T\big(T^{d-1} + a_{d-1}T^{d-2} + \cdots + a_1 I\big) = -a_0 I.$$

So $S = -\frac{1}{a_0}\big(T^{d-1} + a_{d-1}T^{d-2} + \cdots + a_1 I\big)$ satisfies $TS = I$, and $ST = I$ because S is a polynomial in T. Hence T is invertible with $T^{-1} = S$.

b) Write $m_T = xq$ with q monic of degree $d-1$. By minimality $q(T) \neq 0$ (if $d = 1$, $q = 1$ and $q(T) = I \neq 0$ since $V \neq \{0\}$). Pick v with $w = q(T)v \neq 0$. Then $Tw = Tq(T)v = m_T(T)v = 0$, so T is not injective, hence not invertible.

c) $R^2(x,y) = R(-y,x) = (-x,-y)$, so $R^2 + I = 0$ and m_R divides x^2+1. m_R is not of degree 1: $R = cI$ is impossible since $R(1,0) = (0,1)$ is not a multiple of $(1,0)$. So $m_R = x^2 + 1$.

d) $d = 2$, $a_1 = 0$, $a_0 = 1$: $R^{-1} = -\frac{1}{1}(R + 0\cdot I) = -R$, i.e. $R^{-1}(x,y) = (y,-x)$. Check: $R(y,-x) = (x,y)$.

§2 Additional midterm practice: an operator with $T^2 = T$

(a) Let $w \in \operatorname{range} T \cap \operatorname{null} T$, say $w = Tu$. Then $w = Tu = T^2u = T(Tu) = Tw = 0$.

(b) For $v \in V$, $v = Tv + (v - Tv)$ with $Tv \in \operatorname{range} T$ and $T(v - Tv) = Tv - T^2v = 0$, so $v - Tv \in \operatorname{null} T$.

(c) If $w = Tu \in \operatorname{range} T$, then $Sw = STu = T(Su) \in \operatorname{range} T$. If $v \in \operatorname{null} T$, then $T(Sv) = S(Tv) = S0 = 0$, so $Sv \in \operatorname{null} T$. By (a) and (b), $V = \operatorname{range} T \oplus \operatorname{null} T$. Take a basis $u_1, \ldots, u_k$ of $\operatorname{range} T$ and $w_1, \ldots, w_l$ of $\operatorname{null} T$;

together they form a basis of V. Each Su_i is a combination of the u's and each Sw_j is a combination of the w's, so

$$\mathcal{M}(S) = \begin{pmatrix} A & 0 \\ 0 & B \end{pmatrix}, \qquad A = \mathcal{M}(S|_{\operatorname{range} T}),\ B = \mathcal{M}(S|_{\operatorname{null} T}).$$

§3 Tutoring program review exercises

1. Dual space practice.
Write $f_1 = 1$, $f_2 = \sin x$, $f_3 = \sin 2x$.

(a) Values on the basis: $\delta_0 = (1,0,0)$, $\delta_{\pi/2} = (1,1,0)$, $\delta_0' = (0,1,2)$ (as $f_j \mapsto$ value; $\sin\pi = 0$, $\cos 0 = 1$, $\frac{d}{dx}\sin 2x|_0 = 2$). The matrix $M = \left(\begin{smallmatrix} 1&0&0\\1&1&0\\0&1&2\end{smallmatrix}\right)$ with these rows is invertible (lower-triangular, diagonal $1,1,2$), so the three functionals are independent. Since $\dim V' = 3$, they form a basis.

(b) The dual basis is given by the rows of $M^{-1} = \left(\begin{smallmatrix} 1&0&0\\-1&1&0\\ \frac12&-\frac12&\frac12\end{smallmatrix}\right)$ in terms of $\delta_0, \delta_{\pi/2}, \delta_0'$:

$$\varphi_1 = \delta_0, \quad \varphi_2 = -\delta_0 + \delta_{\pi/2}, \quad \varphi_3 = \tfrac12\big(\delta_0 - \delta_{\pi/2} + \delta_0'\big).$$

Check: $\varphi_3(1) = \frac12(1-1+0) = 0$, $\varphi_3(\sin x) = \frac12(0-1+1) = 0$, $\varphi_3(\sin 2x) = \frac12(0-0+2) = 1$.

(c) $\varphi(1) = 1$, $\varphi(\sin x) = \frac{1}{\pi}[-\cos x]_0^\pi = \frac{2}{\pi}$, $\varphi(\sin 2x) = \frac{1}{\pi}\big[-\frac12\cos 2x\big]_0^\pi = 0$. So $\varphi = \varphi_1 + \frac{2}{\pi}\varphi_2 = \big(1 - \frac{2}{\pi}\big)\delta_0 + \frac{2}{\pi}\delta_{\pi/2}$.

2. Duals of the inclusion and the quotient map.

(a) $i'(\varphi) = \varphi \circ i = \varphi|_U$. $\operatorname{null} i' = (\operatorname{range} i)^0 = U^0$; $\operatorname{range} i' = (\operatorname{null} i)^0 = \{0\}^0 = U'$ (i injective, so i' surjective). $\operatorname{null}\pi' = (\operatorname{range}\pi)^0 = (V/U)^0 = \{0\}$ (π surjective, so π' injective); $\operatorname{range}\pi' = (\operatorname{null}\pi)^0 = U^0$.

(b) $\operatorname{range}\pi' = U^0$. Define $T\colon U' \to V'/U^0$ by $T(\psi) = \tilde\psi + U^0$, where $\tilde\psi \in V'$ is any extension of ψ (extend a basis of U to V and let $\tilde\psi$ be 0 on the new vectors). Well-defined: two extensions differ by a functional that is 0 on U, i.e. an element of U^0. Linear: $\tilde\psi_1 + \tilde\psi_2$ extends $\psi_1 + \psi_2$. Injective: $\tilde\psi \in U^0$ gives $\psi = \tilde\psi|_U = 0$. Surjective: $\varphi + U^0 = T(\varphi|_U)$. Equivalently, $i'\colon V' \to U'$ is surjective with null space U^0, and T is the inverse of the isomorphism $V'/U^0 \to U'$ from the first isomorphism theorem.

3. $p(T)v = p(\lambda)v$.
By induction $T^k v = T(T^{k-1}v) = T(\lambda^{k-1}v) = \lambda^k v$. So for $p = \sum a_k x^k$, $p(T)v = \sum a_k T^k v = \sum a_k \lambda^k v = p(\lambda)v$.

4. Differentiation and rotations.

1. $d(1) = 0$, $d(\sin x) = \cos x$, $d(\cos x) = -\sin x$, $d(\sin 2x) = 2\cos 2x$, $d(\cos 2x) = -2\sin 2x$. If $df = \lambda f$ with $\lambda \in \mathbb{R}$ and $f = a + b\sin x + c\cos x + e\sin 2x + g\cos 2x$, comparing coefficients gives $\lambda a = 0$, $\lambda b = -c$, $\lambda c = b$, $\lambda e = -2g$, $\lambda g = 2e$. Then $(\lambda^2+1)b = 0$ and $(\lambda^2+4)e = 0$, so $b = c = e = g = 0$, $f = a$, and $\lambda = 0$. The only eigenvalue is 0, with $E(0,d) =$ constants, of dimension $1 < 5$, so d is not diagonalizable over $\mathbb{R}$. ($m_d = x(x^2+1)(x^2+4)$; over $\mathbb{C}$ the eigenvalues would be $0, \pm i, \pm 2i$ and d would be diagonalizable.)
2. $T^2 = -I$, so the eigenvalues are roots of $\lambda^2 + 1$: $\pm i$. $T(x,y) = i(x,y)$ means $y = ix$: eigenvector $(1,i)$. In the basis $(1,i), (0,1)$: $T(1,i) = i(1,i)$ and $T(0,1) = (1,0) = 1\cdot(1,i) - i\cdot(0,1)$, so the matrix is $\left(\begin{smallmatrix} i & 1\\ 0 & -i\end{smallmatrix}\right)$. (The eigenbasis $(1,i), (1,-i)$ even gives $\operatorname{diag}(i,-i)$.)
3. If the matrix in a basis v_1, v_2 were upper-triangular, then $Tv_1 \in \operatorname{span}(v_1)$, so v_1 would be a real eigenvector. But $(y,-x) = \lambda(x,y)$ gives $x = -\lambda y = -\lambda^2 x$, so $(1+\lambda^2)x = 0$, $x = 0$, then $y = 0$. Contradiction.

5. A 3×3 matrix.
The matrix is upper-triangular, so the eigenvalues are 3 and 1, and the characteristic polynomial is $(x-3)^2(x-1)$. $E(3) = \operatorname{null}\left(\begin{smallmatrix} 0&1&1\\0&0&-1\\0&0&-2\end{smallmatrix}\right)$: $z = 0$, $y = 0$, so $E(3) = \operatorname{span}(e_1)$. $E(1) = \operatorname{null}\left(\begin{smallmatrix} 2&1&1\\0&2&-1\\0&0&0\end{smallmatrix}\right)$: $z = 2y$, $2x = -3y$, so $E(1) = \operatorname{span}\big((-3,2,4)\big)$. Minimal polynomial: its roots are exactly $1, 3$ and $\deg m_A \le 3$, so m_A is $(x-3)(x-1)$, $(x-3)^2(x-1)$ or $(x-3)(x-1)^2$. $(A-3I)(A-I)$ has first row $(0,2,-1) \neq 0$ and $(A-3I)(A-I)^2$ has first row $(0,4,-2) \neq 0$, while $(A-3I)^2 = \left(\begin{smallmatrix} 0&0&-3\\0&0&2\\0&0&4\end{smallmatrix}\right)$ gives $(A-3I)^2(A-I) = 0$ (the third row of $A-I$ is 0). So $m_A = (x-3)^2(x-1)$. Not diagonalizable.

6. The polynomial of a single vector.

(a) With $n = \dim V$, the $n+1$ vectors $v, Tv, \dots, T^n v$ are dependent, so some nonzero polynomial q has $q(T)v = 0$. Among all such, take one of smallest degree and divide by its leading coefficient: this is p_v. If p and $\tilde{p}$ are two such, both monic of the smallest degree, then $(p-\tilde{p})(T)v = 0$ and $\deg(p-\tilde{p})$ is smaller; so $p - \tilde{p} = 0$.

(b) Divide: $m_T = p_v r + s$ with $\deg s < \deg p_v$. Then $s(T)v = m_T(T)v - r(T)p_v(T)v = 0 - r(T)0 = 0$. By minimality of $\deg p_v$, $s = 0$, so p_v divides m_T.

7. Invariant subspaces and annihilators.
If $v \in \operatorname{null} T$, then $T(Tv) = T0 = 0$, so $Tv \in \operatorname{null} T$. If $w = Tu \in \operatorname{range} T$, then $Tw = T(Tu) \in \operatorname{range} T$. In general, if U is T-invariant and $\varphi \in U^0$, then $(T'\varphi)(u) = \varphi(Tu) = 0$ for $u \in U$ (since $Tu \in U$), so $T'\varphi \in U^0$. Apply this to $U = \operatorname{null} T$ and $U = \operatorname{range} T$.

8. True or false.

1. False: e_1, e_2 for $\operatorname{diag}(1, 2)$; also $v_2 = -v_1$ gives 0.
2. False: T injective iff T' surjective. The inclusion $\mathbb{F} \to \mathbb{F}^2$ is injective but its dual is not.
3. True: $T = I$ gives $V = E(1, I)$.
4. False: $T = I$, $S = -I$: $\operatorname{null}(T+S) = V$ but $\operatorname{null} T + \operatorname{null} S = \{0\}$.
5. False: rotation by 90° on $\mathbb{R}^2$. (True for odd-dimensional real spaces.)
6. False: I on $\mathbb{F}^2$ has $\dim E(1) = 2$ but $m = x - 1$; the shear has $\dim E(1) = 1$ but $m = (x-1)^2$.
7. True: $\left(\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right)$ has $m = x - 1$ and $\left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$ has $m = (x-1)^2$.

§4 Homework problems

HW4 P1.

(a) $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)\left(\begin{smallmatrix} d & -b \\ -c & a \end{smallmatrix}\right) = \left(\begin{smallmatrix} ad-bc & -ab+ba \\ cd-dc & -cb+da \end{smallmatrix}\right) = (ad-bc)I$.

(b) Let $B = \frac{1}{ad-bc}\left(\begin{smallmatrix} d & -b \\ -c & a \end{smallmatrix}\right)$. By (a), $AB = I$. Also $\left(\begin{smallmatrix} d & -b \\ -c & a \end{smallmatrix}\right)\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) = \left(\begin{smallmatrix} da-bc & db-bd \\ -ca+ac & -cb+ad \end{smallmatrix}\right) = (ad-bc)I$, so $BA = I$. Hence $A^{-1} = \frac{1}{ad-bc}\left(\begin{smallmatrix} d & -b \\ -c & a \end{smallmatrix}\right)$.

(c) Let $M = \left(\begin{smallmatrix} d & -b \\ -c & a \end{smallmatrix}\right)$; by (a), $AM = 0$. If $M \neq 0$, some column m of M is nonzero and $Am = 0$, so A is not injective. If $M = 0$, then $a = b = c = d = 0$ and $A = 0$. Either way A is not invertible.

HW4 P2.
$\det\left(\begin{smallmatrix} 2 & 1 \\ 5 & 3 \end{smallmatrix}\right) = 6 - 5 = 1$, inverse $\left(\begin{smallmatrix} 3 & -1 \\ -5 & 2 \end{smallmatrix}\right)$. $\det\left(\begin{smallmatrix} 1 & i \\ i & -1 \end{smallmatrix}\right) = -1 - i^2 = 0$: not invertible. Over $\mathbb{F}_5$, $\det\left(\begin{smallmatrix} 1 & 2 \\ 3 & 1 \end{smallmatrix}\right) = 1 - 6 = -5 = 0$: not invertible (the second column is 2 times the first, since $6 = 1$ in $\mathbb{F}_5$).

HW4 P3.
(a)⇒(b): if $\sum c_i T v_i = 0$, then $T\left(\sum c_i v_i\right) = 0$; T is injective, so $\sum c_i v_i = 0$ and all $c_i = 0$. So $Tv_1, \dots, Tv_n$ is an independent list of $n = \dim V$ vectors, hence a basis. (b)⇒(a): take any basis $v_1, \dots, v_n$. Then $\operatorname{range} T \supseteq \operatorname{span}(Tv_1, \dots, Tv_n) = V$, so T is surjective, hence (operator on a finite-dimensional space) invertible.

HW4 P4.

(a) View B and C as linear maps ($x \mapsto Bx$). $\operatorname{range}(BC) = B(\operatorname{range} C) \subseteq \operatorname{range} B$, so $\operatorname{rank} BC \leq \operatorname{rank} B$. Also $B(\operatorname{range} C)$ is the range of B restricted to $\operatorname{range} C$; by rank–nullity its dimension is at most $\dim \operatorname{range} C$, so $\operatorname{rank} BC \leq \operatorname{rank} C$.

(b) $B = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right)$, $C = \left(\begin{smallmatrix} 0 & 0 \\ 0 & 1 \end{smallmatrix}\right)$: both have rank 1, and $BC = 0$ has rank 0.

HW4 P6.

(a) If $v + U$ is a subspace, it contains 0, so $0 = v + u$ for some $u \in U$, and $v = -u \in U$. Conversely, if $v \in U$: $v + u \in U$ for $u \in U$, so $v + U \subseteq U$; and $u = v + (u - v)$ with $u - v \in U$, so $U \subseteq v + U$. Hence $v + U = U$, a subspace.

(b) If $v - v' \in U$, then $v + u = v' + \big((v - v') + u\big) \in v' + U$, so $v + U \subseteq v' + U$; since $v' - v \in U$ too, the reverse inclusion holds. Conversely, $v = v + 0 \in v + U = v' + U$, so $v = v' + u$ for some $u \in U$, and $v - v' = u \in U$.

HW4 P7.
Let $u_1, \dots, u_m$ be a basis of U. Since S is injective, $Su_1, \dots, Su_m$ is independent in V. Extend $u_1, \dots, u_m$ to a basis $u_1, \dots, u_m, w_1, \dots, w_k$ of V, and $Su_1, \dots, Su_m$ to a basis $Su_1, \dots, Su_m, y_1, \dots, y_k$ of V (both have $\dim V = m + k$ vectors). Define T by $Tu_i = Su_i$ and $Tw_j = y_j$. Then $T|_U = S$ (they agree on a basis of U), and $\operatorname{range} T$ contains the basis $Su_1, \dots, Su_m, y_1, \dots, y_k$ of V, so T is surjective, hence invertible (an operator on a finite-dimensional space).

HW5 P4(a).
Write $v = \sum_k a_k v_k$. Then $\varphi_j(v) = \sum_k a_k \varphi_j(v_k) = a_j$, so $v = \sum_j \varphi_j(v) v_j$.

HW5 P6.
(a) $(a,b,c) \in U^0 \iff a + b = 0$ and $b + c = 0 \iff (a,b,c) = a(1,-1,1)$. So $U^0 = \operatorname{span}\big((1,-1,1)\big)$. (b) $(1,1,0), (0,1,1)$ are independent, so $\dim U = 2$; $\dim U^0 = 1$; $2 + 1 = 3$.

HW6 P2.

(a) If $Dp = \lambda p$ with $\lambda \neq 0$ and $p \neq 0$, then $\deg(\lambda p) = \deg p$ but $\deg Dp < \deg p$ (or $Dp = 0$), a contradiction. So the only eigenvalue is 0, and $E(0, D) = \operatorname{null} D = \mathcal{P}_0(\mathbb{F})$, the constants. (Equivalently: $\mathcal{M}(D)$ in $1, x, \dots, x^n$ is upper-triangular with zero diagonal.)

(b) On smooth functions $\mathbb{R} \to \mathbb{F}$, $\frac{d}{dx} e^{\lambda x} = \lambda e^{\lambda x}$ with $e^{\lambda x} \neq 0$, so every $\lambda \in \mathbb{F}$ is an eigenvalue (every real λ for $\mathbb{F} = \mathbb{R}$, every complex λ for $\mathbb{F} = \mathbb{C}$). If $f' = \lambda f$, then $(e^{-\lambda x} f)' = e^{-\lambda x}(f' - \lambda f) = 0$, so $f = c e^{\lambda x}$; hence $E(\lambda, D) = \operatorname{span}(e^{\lambda x})$. The degree argument fails because $e^{\lambda x}$ is not a polynomial; $\mathcal{P}_n$ is finite-dimensional and D lowers degree there.

HW6 P4.
$tI - A = \left(\begin{smallmatrix} t-a & -b \\ -c & t-d \end{smallmatrix}\right)$, so $\det(tI - A) = (t-a)(t-d) - (-b)(-c) = t^2 - (a+d)t + (ad - bc)$. Then: λ is an eigenvalue $\iff$ $\lambda I - A$ is not injective $\iff$ it is not invertible $\iff$ $\det(\lambda I - A) = 0$ (HW4 P1) $\iff$ λ is a root.

HW6 P5.
A_1: $\det(tI - A_1) = t^2 + 1$. Over $\mathbb{R}$: no eigenvalues, no eigenbasis. Over $\mathbb{C}$: $\pm i$; $E(i) = \operatorname{span}\big((1,-i)\big)$, $E(-i) = \operatorname{span}\big((1,i)\big)$; these form an eigenbasis. A_2: $(t-1)^2$; $E(1) = \operatorname{null}\left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right) = \operatorname{span}(e_1)$, dimension $1 < 2$: no eigenbasis. A_3: $t^2 - t - 1$, roots $\lambda_\pm = \frac{1 \pm \sqrt{5}}{2}$. $A_3(1, \lambda) = (\lambda, 1 + \lambda) = \lambda(1, \lambda)$ because $\lambda^2 = \lambda + 1$, so $(1, \lambda_\pm) \in E(\lambda_\pm)$; since $A_3 \neq \lambda_\pm I$, $\dim E(\lambda_\pm) \leq 1$, so $E(\lambda_\pm) = \operatorname{span}\big((1, \lambda_\pm)\big)$. Two distinct real eigenvalues: eigenbasis.

HW6 P6.

(a) $T\big(\sum_k c_k T^k v\big) = \sum_k c_k T^{k+1} v \in C(v, T)$.

(b) The $\dim V + 1$ vectors $v, \dots, T^{\dim V} v$ are dependent, so there is a smallest m with $T^m v \in \operatorname{span}(v, \dots, T^{m-1} v)$, and $m \geq 1$ since $v \neq 0$. The list $v, \dots, T^{m-1} v$ is independent: in a dependence, the last nonzero coefficient at $T^j v$ with $j < m$ would give $T^j v \in \operatorname{span}(v, \dots, T^{j-1} v)$, contradicting minimality. So $m \leq \dim V$. $W = \operatorname{span}(v, \dots, T^{m-1} v)$ is T-invariant: $T(T^k v) = T^{k+1} v \in W$ for $k < m - 1$, and $T(T^{m-1} v) = T^m v \in W$. By induction every $T^k v \in W$, so $C(v, T) \subseteq W \subseteq C(v, T)$.

HW6 P7.
$Pp = p$ gives $p_1 = \frac{1}{2} p_1 + \frac{1}{2} p_2$, so $p_1 = p_2$, and $p_2 = \frac{1}{2} p_1 + \frac{1}{2} p_3$, so $p_3 = p_2$. With $p_1 + p_2 + p_3 = 1$: $p = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$. Column j of P lists the probabilities of moving from state j to each state. If the chain's distribution is p at one step, it is p at the next: p is a stationary distribution. Also $P(1, 0, -1) = \frac{1}{2}(1, 0, -1)$ and $P(1, -2, 1) = -\frac{1}{2}(1, -2, 1)$, so $p, (1, 0, -1), (1, -2, 1)$ is an eigenbasis. A probability vector x is $x = p + a(1, 0, -1) + b(1, -2, 1)$ (the coefficient of p is 1 because the other two vectors have entry sum 0 while x and p have entry sum 1), so $P^k x = p + a 2^{-k}(1, 0, -1) + b(-\frac{1}{2})^k (1, -2, 1) \to p$: whatever the starting distribution, in the long run the chain is in each state with probability $\frac{1}{3}$.

HW7 P1.

(a) $A^2 = -I$, so m_A divides $x^2 + 1$; A is not a multiple of I, so $m_A = x^2 + 1$ over $\mathbb{R}$. The definition (first k with $A^k \in \operatorname{span}(I, \dots, A^{k-1})$) gives the same polynomial over $\mathbb{C}$: $x^2 + 1 = (x - i)(x + i)$.

(b) $(B - I)^2 = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)^2 = 0$ and $B \neq I$, so $m_B = (x - 1)^2$.

(c) A over $\mathbb{R}$: $x^2 + 1$ is not a product of real linear factors, so A is not diagonalizable. Over $\mathbb{C}$: distinct roots $\pm i$, diagonalizable. B: repeated root, not diagonalizable. This matches HW6 P5.

HW7 P2.
(a) D lowers degree by 1, so $D^4 = 0$ on $\mathcal{P}_3(\mathbb{R})$, while $D^3x^3 = 6 \neq 0$. m_D divides x^4 and is not x^k for $k \leq 3$, so $m_D = x^4$. (b) Repeated root, so not diagonalizable (also: $E(0, D)$ is the constants, of dimension $1 < 4$).

HW7 P3(c).
Since $m_T(T) = 0$, $T^4 + T^3 - T^2 - 5T + 4I = 0$, so $T(T^3 + T^2 - T - 5I) = -4I$ and $T^{-1} = -\frac{1}{4}(T^3 + T^2 - T - 5I)$. $m_T = (x-1)^2(x^2+3x+4)$, so the eigenvalues are 1 and $\frac{-3\pm i\sqrt{7}}{2}$. 1 is a repeated root, so T is not diagonalizable.

HW7 P4(b).
Let $p(x) = c^d m_T(x/c)$; it is monic of degree d (leading coefficient $c^d \cdot c^{-d}$). Then $p(cT) = c^d m_T(T) = 0$, so m_{cT} divides p and $\deg m_{cT} \leq d$. Applying the same argument to cT and $1/c$ gives $d = \deg m_T \leq \deg m_{cT}$. So m_{cT} divides p, they have the same degree and both are monic: $m_{cT} = p$.

HW7 P6.
Upper-triangular: eigenvalues 2 and 3. $E(2) = \operatorname{null}\left(\begin{smallmatrix}0&1&0\\0&0&0\\0&0&1\end{smallmatrix}\right) = \operatorname{span}(e_1)$, $E(3) = \operatorname{null}\left(\begin{smallmatrix}-1&1&0\\0&-1&0\\0&0&0\end{smallmatrix}\right) = \operatorname{span}(e_3)$. m_C has roots exactly $2, 3$ and degree ≤ 3: candidates $(x-2)(x-3)$, $(x-2)^2(x-3)$, $(x-2)(x-3)^2$. $(C-2I)(C-3I)$ and $(C-2I)(C-3I)^2$ both have $(1,2)$ entry $\mp 1 \neq 0$, while $(C-2I)^2 = \operatorname{diag}(0,0,1)$ and $(C-2I)^2(C-3I) = 0$. So $m_C = (x-2)^2(x-3)$: repeated root, not diagonalizable ($\dim E(2) + \dim E(3) = 2 < 3$).

HW7 P7.
$p_j(x) = \prod_{k\neq j} \frac{x-\lambda_k}{\lambda_j-\lambda_k}$, so $p_j(\lambda_j) = 1$ and $p_j(\lambda_k) = 0$ for $k \neq j$. Since m_T has distinct roots, T is diagonalizable and $V = \bigoplus_k E(\lambda_k, T)$. For $v \in E(\lambda_k, T)$, $p_j(T)v = p_j(\lambda_k)v$, which is v if $k = j$ and 0 otherwise. So writing $v = \sum_k v_k$ with $v_k \in E(\lambda_k, T)$, $p_j(T)v = v_j$: this is the projection onto $E(\lambda_j, T)$ along $\bigoplus_{k\neq j} E(\lambda_k, T)$.

§5 Practice Set 1

1. The second isomorphism theorem.

a) $T(u_1 + u_2) = (u_1 + u_2) + W = (u_1 + W) + (u_2 + W)$ and $T(cu) = c(u + W)$. Every element of $(U + W)/W$ is $(u + w) + W$ with $u \in U$, $w \in W$, and $(u + w) + W = u + W = T(u)$ because $(u + w) - u = w \in W$.

b) $T(u) = 0 + W \iff u - 0 \in W \iff u \in W$; since $u \in U$, this means $u \in U \cap W$.

c) First isomorphism theorem: $U/\operatorname{null} T \cong \operatorname{range} T$, i.e. $U/(U \cap W) \cong (U + W)/W$ via $u + (U \cap W) \mapsto u + W$. Dimensions: $\dim U - \dim(U \cap W) = \dim(U + W) - \dim W$.

d) $U \cap W$: upper-triangular and symmetric means the $(2,1)$ and $(1,2)$ entries are 0: the diagonal matrices, dimension 2. $\dim(U + W) = 3 + 3 - 2 = 4$, so $U + W = \mathbb{R}^{2,2}$. $U/(U \cap W)$ has dimension 1, basis $E_{12} + (U \cap W)$ ($E_{12} \in U$, not diagonal). Its image is $E_{12} + W$, which is nonzero (as E_{12} is not symmetric) and spans $\mathbb{R}^{2,2}/W$.

2. Dual bases and null spaces of dual maps.

a) $\dim V' = \dim \mathcal{L}(V, \mathbb{F}) = n \cdot 1 = n$, so it suffices to show independence. If $\sum_j \lambda_j \varphi_j = 0$, apply both sides to v_k: $\sum_j \lambda_j \varphi_j(v_k) = \lambda_k = 0$ for each k.

b) Both sides are linear functionals. On v_k the right side gives $\sum_j \psi(v_j)\varphi_j(v_k) = \psi(v_k)$. Linear maps that agree on a basis are equal.

c) $\varphi \in \operatorname{null} T' \iff \varphi \circ T = 0 \iff \varphi(Tv) = 0$ for all $v \iff \varphi$ vanishes on $\operatorname{range} T \iff \varphi \in (\operatorname{range} T)^0$. Then $\dim \operatorname{null} T' = \dim(\operatorname{range} T)^0 = \dim W - \dim \operatorname{range} T$.

d) The dual basis is the rows of $\left(\begin{smallmatrix}1&1\\1&2\end{smallmatrix}\right)^{-1} = \left(\begin{smallmatrix}2&-1\\-1&1\end{smallmatrix}\right)$: $\varphi_1 = 2x - y$, $\varphi_2 = -x + y$. $\psi = \psi(v_1)\varphi_1 + \psi(v_2)\varphi_2 = 2\varphi_1 + \varphi_2$. Check: $2(2x - y) + (-x + y) = 3x - y$.

3. Eigenvectors and diagonalizability.

a) Suppose the list is dependent and let k be the smallest index with $v_k \in \operatorname{span}(v_1, \ldots, v_{k-1})$; write $v_k = a_1v_1 + \cdots + a_{k-1}v_{k-1}$. Apply T: $\lambda_k v_k = \sum_{i<k} a_i\lambda_i v_i$. Multiply the first equation by λ_k and subtract: $0 = \sum_{i<k} a_i(\lambda_i - \lambda_k)v_i$. By minimality of k, $v_1, \ldots, v_{k-1}$ is independent, so $a_i(\lambda_i - \lambda_k) = 0$, and $\lambda_i \neq \lambda_k$ gives $a_i = 0$. Then $v_k = 0$, but eigenvectors are nonzero. (For $k = 1$ the condition reads $v_1 = 0$, also impossible.)

b) Suppose $u_1 + \cdots + u_m = 0$ with $u_i \in E(\lambda_i, T)$. The nonzero u_i are eigenvectors with distinct eigenvalues whose sum is 0, a dependence with all coefficients 1; by (a) there are no nonzero u_i. So the only way to write 0 is with all $u_i = 0$.

c) If T is diagonalizable, it has an eigenbasis; each basis vector lies in some $E(\lambda_i, T)$, so $V = \sum E(\lambda_i, T)$, which is direct by (b), and $\dim V = \sum \dim E(\lambda_i, T)$. Conversely, the direct sum $\bigoplus E(\lambda_i, T)$ has dimension $\sum \dim E(\lambda_i, T) = \dim V$, so it is V, and the union of bases of the $E(\lambda_i, T)$ is a basis of V made of eigenvectors.

d) Upper-triangular: eigenvalues 1 (twice) and 2. $E(1) = \operatorname{null}\left(\begin{smallmatrix} 0 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{smallmatrix}\right) = \{y + z = 0\} = \operatorname{span}\big((1,0,0),(0,1,-1)\big)$, dimension 2. $E(2) = \operatorname{null}\left(\begin{smallmatrix} -1 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & -1 \end{smallmatrix}\right) = \operatorname{span}\big((1,1,0)\big)$. $2 + 1 = 3$, so A is diagonalizable, with eigenbasis $(1,0,0),(0,1,-1),(1,1,0)$.

4. Diagonalizability and the minimal polynomial.

a) Let $p = (x - \lambda_1)\cdots(x - \lambda_m)$. For $v \in E(\lambda_i, T)$, the factors of $p(T)$ commute, so $p(T)v = \big(\prod_{j\neq i}(T - \lambda_j I)\big)(T - \lambda_i I)v = 0$. As $V = \bigoplus E(\lambda_i, T)$, $p(T) = 0$, so m_T divides p. Each λ_i is an eigenvalue, hence a zero of m_T, so each $x - \lambda_i$ divides m_T; since the λ_i are distinct, p divides m_T. Both are monic, so $m_T = p$.

b) Evaluating $g = (x - \lambda_1)q + s$ at λ_1 gives $s = g(\lambda_1) = \prod_{j\geq 2}(\lambda_1 - \lambda_j) \neq 0$. For $v \in V$ let $u = \frac{1}{s}g(T)v$. Then $(T - \lambda_1 I)u = \frac{1}{s}m_T(T)v = 0$, so $u \in \operatorname{null}(T - \lambda_1 I)$. And

$$v - u = \tfrac{1}{s}\big(sv - g(T)v\big) = -\tfrac{1}{s}q(T)(T - \lambda_1 I)v = (T - \lambda_1 I)\big(-\tfrac{1}{s}q(T)v\big) \in \operatorname{range}(T - \lambda_1 I).$$

So $v = u + (v - u)$ shows $V = \operatorname{null}(T - \lambda_1 I) + \operatorname{range}(T - \lambda_1 I)$.

c) By rank–nullity, $\dim\operatorname{null} + \dim\operatorname{range} = \dim V = \dim(\operatorname{null} + \operatorname{range})$, so the intersection has dimension 0 and the sum is direct. U is invariant: $T(T - \lambda_1 I)w = (T - \lambda_1 I)Tw$. $m_T(T|_U) = m_T(T)|_U = 0$, so $m_{T|_U}$ divides m_T, hence is a product of distinct linear factors. λ_1 is an eigenvalue, so $\operatorname{null}(T - \lambda_1 I) \neq \{0\}$ and $\dim U < \dim V$. By induction on $\dim V$ (base case $\dim V \leq 1$, and $m = 1$ means $T = \lambda_1 I$), $T|_U$ has an eigenbasis. Together with a basis of $\operatorname{null}(T - \lambda_1 I)$, this is an eigenbasis of $V = \operatorname{null}(T - \lambda_1 I) \oplus U$.

d) $T - 2I = D$ (differentiation). On $\mathcal{P}_2(\mathbb{R})$, $D^3 = 0$ and $D^2 \neq 0$ ($D^2 x^2 = 2$). So $(T - 2I)^3 = 0$, m_T divides $(x-2)^3$, and m_T is not $(x - 2)$ or $(x - 2)^2$. Hence $m_T = (x - 2)^3$, which has a repeated root: T is not diagonalizable. (Matrix in $1, x, x^2$: $\left(\begin{smallmatrix} 2 & 1 & 0 \\ 0 & 2 & 2 \\ 0 & 0 & 2 \end{smallmatrix}\right)$.)

§6 Practice Set 2

1. Isomorphisms and products.

a) If $Tv = 0$, then $v = STv = 0$, so T is injective. An injective operator on a finite-dimensional space is invertible. Then $S = S(TT^{-1}) = (ST)T^{-1} = T^{-1}$, so $TS = TT^{-1} = I$.

b) $DJp = \frac{d}{dx}\int_0^x p(t)\,dt = p$ by the fundamental theorem of calculus. $JDp = \int_0^x p'(t)\,dt = p(x) - p(0)$, so $JD(1) = 0 \neq 1$.

c) Span: $(v, w) = (v, 0) + (0, w) = \sum a_i(v_i, 0) + \sum b_j(0, w_j)$ where $v = \sum a_i v_i$, $w = \sum b_j w_j$. Independent: $\sum a_i(v_i, 0) + \sum b_j(0, w_j) = \big(\sum a_i v_i, \sum b_j w_j\big) = (0, 0)$ forces $\sum a_i v_i = 0$ and $\sum b_j w_j = 0$, so all a_i, b_j are 0.

d) $\Gamma(u_1, u_2) = 0 \iff u_1 = -u_2$; then $u_1 \in U_1$ and $u_1 = -u_2 \in U_2$, so $(u_1, u_2) = (u, -u)$ with $u \in U_1 \cap U_2$; conversely each such pair is in $\operatorname{null}\Gamma$. Γ maps onto $U_1 + U_2$, so rank–nullity gives $\dim(U_1 + U_2) = \dim U_1 + \dim U_2 - \dim\operatorname{null}\Gamma$, and $\dim\operatorname{null}\Gamma = \dim(U_1 \cap U_2)$ via $u \mapsto (u, -u)$. The sum is direct $\iff U_1 \cap U_2 = \{0\} \iff \dim(U_1 + U_2) = \dim U_1 + \dim U_2$.

2. Annihilators and the double dual.

a) Take a basis $v_1, \ldots, v_m$ of U and extend to a basis $v_1, \ldots, v_n$ of V, with dual basis $\varphi_1, \ldots, \varphi_n$. For $\varphi = \sum c_j\varphi_j$, $\varphi(v_k) = c_k$, so $\varphi \in U^0 \iff c_1 = \cdots = c_m = 0 \iff \varphi \in \operatorname{span}(\varphi_{m+1}, \ldots, \varphi_n)$. These $n - m$ functionals are independent, so $\dim U^0 = n - m$.

b) If $\varphi = T'\psi$ and $v \in \operatorname{null} T$, then $\varphi(v) = \psi(Tv) = \psi(0) = 0$; so $\operatorname{range} T' \subseteq (\operatorname{null} T)^0$. Dimensions:

$$\begin{aligned}\dim\operatorname{range} T' &= \dim W' - \dim\operatorname{null} T' = \dim W - (\dim W - \dim\operatorname{range} T)\\ &= \dim\operatorname{range} T = \dim V - \dim\operatorname{null} T = \dim(\operatorname{null} T)^0.\end{aligned}$$

A subspace of the same finite dimension is the whole space, so equality holds.

c) $j_V(v)$ is linear in φ, and j_V is linear in v. If $v \neq 0$, extend v to a basis $v, v_2, \ldots, v_n$ and let φ be the first dual basis functional: $\varphi(v) = 1$. Then $j_V(v)(\varphi) = 1$, so $j_V(v) \neq 0$; j_V is injective. Since $\dim V'' = \dim V' = \dim V$, an injective linear map $V \to V''$ is an isomorphism.

d) $a = (a_1, a_2, a_3, a_4) \in U^0 \iff a_1 - a_3 = 0$ and $a_2 - a_4 = 0$. Basis: $(1,0,1,0)$, $(0,1,0,1)$. ($\dim U^0 = 4 - 2$.)

3. Eigenvectors over $\mathbb{C}$ and upper-triangular matrices.

a) $v, Tv, \dots, T^n v$ are $n+1$ vectors in an n-dimensional space, so they are dependent; dropping zero top coefficients gives a relation with $a_m \neq 0$. A relation with $m = 0$ would read $a_0 v = 0$ with $a_0 \neq 0$, impossible as $v \neq 0$. So the smallest m is ≥ 1, and p is nonconstant.

b) By the fundamental theorem of algebra p has a zero $\lambda \in \mathbb{C}$, so $p(z) = (z-\lambda)q(z)$ with $\deg q = m-1$ and leading coefficient $a_m \neq 0$. If $w = q(T)v$ were 0, it would be a nontrivial relation among $v, \dots, T^{m-1}v$, contradicting minimality. So $w \neq 0$, and $(T-\lambda I)w = (T-\lambda I)q(T)v = p(T)v = 0$: $Tw = \lambda w$.

c) Let $V_j = \operatorname{span}(v_1, \dots, v_j)$, $V_0 = \{0\}$. The matrix of $T - \lambda_k I$ is upper-triangular with (k,k) entry 0, so $(T-\lambda_k I)v_j \in V_j \subseteq V_{k-1}$ for $j < k$ and $(T-\lambda_k I)v_k \in V_{k-1}$. So $T - \lambda_k I$ maps the k-dimensional V_k into the $(k-1)$-dimensional V_{k-1}, and cannot be injective on V_k: some $w \neq 0$ has $Tw = \lambda_k w$.

d) Matrix $\left(\begin{smallmatrix} 1 & -1 \\ 1 & 3 \end{smallmatrix}\right)$; $\lambda^2 - 4\lambda + 4 = (\lambda-2)^2$, so the only eigenvalue is 2, and $E(2) = \operatorname{null}\left(\begin{smallmatrix} -1 & -1 \\ 1 & 1 \end{smallmatrix}\right) = \operatorname{span}\big((1,-1)\big)$. Basis $v_1 = (1,-1)$, $v_2 = (1,0)$: $Tv_1 = 2v_1$, $Tv_2 = (1,1) = -v_1 + 2v_2$. Matrix $\left(\begin{smallmatrix} 2 & -1 \\ 0 & 2 \end{smallmatrix}\right)$. Not diagonalizable: $\dim E(2) = 1 < 2$.

4. Basic properties of the minimal polynomial.

a) If $q = m_T r$, then $q(T) = m_T(T)r(T) = 0$. Conversely, divide: $q = m_T r + s$ with $s = 0$ or $\deg s < \deg m_T$. Then $s(T) = q(T) - m_T(T)r(T) = 0$. If $s \neq 0$, dividing s by its leading coefficient gives a monic polynomial of smaller degree than m_T that sends T to 0, which is impossible. So $s = 0$.

b) If $Tv = \lambda v$, $v \neq 0$, then $0 = m_T(T)v = m_T(\lambda)v$, so $m_T(\lambda) = 0$. Conversely, if $m_T(\lambda) = 0$, write $m_T = (x-\lambda)q$. Then $0 = m_T(T) = q(T)(T-\lambda I)$. If λ were not an eigenvalue, $T - \lambda I$ would be invertible and $q(T) = 0$, with q monic of smaller degree, contradicting minimality.

c) Let $p(x) = m_T(x+\lambda)$, monic of degree $d = \deg m_T$. Then $p(T-\lambda I) = m_T(T - \lambda I + \lambda I) = m_T(T) = 0$, so $m_{T-\lambda I}$ divides p. Similarly $r(x) = m_{T-\lambda I}(x-\lambda)$ is monic with $r(T) = m_{T-\lambda I}(T-\lambda I) = 0$, so $d \leq \deg m_{T-\lambda I}$. Hence $m_{T-\lambda I}$ divides p and has the same degree; both are monic, so they are equal.

d) $\deg m_A \leq 2$ and its zeros are the eigenvalues, so over $\mathbb{C}$ it splits with only the root λ: $m_A = x - \lambda$ or $(x-\lambda)^2$. $m_A = x - \lambda$ exactly when $A - \lambda I = 0$. A is diagonalizable iff m_A has no repeated root iff $m_A = x - \lambda$ iff $A = \lambda I$.

§7 Practice Set 3

1. Induced maps and the third isomorphism theorem.

a) Define $\tilde{T}(v+U) = Tv$. Well-defined: if $v + U = v' + U$ then $v - v' \in U \subseteq \operatorname{null} T$, so $Tv = Tv'$. Linear: $\tilde{T}\big((v+U) + (w+U)\big) = \tilde{T}\big((v+w)+U\big) = Tv + Tw$, and similarly for scalars. Unique: any such map is prescribed on every element $v+U$ of V/U. $\tilde{T}(v+U) = 0 \iff Tv = 0 \iff v \in \operatorname{null} T$, so $\operatorname{null}\tilde{T} = \{v+U : v \in \operatorname{null} T\} = (\operatorname{null} T)/U$.

b) Apply (a) to the quotient map $\pi_U \colon V \to V/U$, whose null space U contains W: we get $\tilde{\pi} \colon V/W \to V/U$, $v + W \mapsto v + U$. It is surjective, since $v + U = \tilde{\pi}(v+W)$, and $\operatorname{null}\tilde{\pi} = \{v+W : v \in U\} = U/W$. By the first isomorphism theorem, $(V/W)/(U/W) \cong V/U$.

c) $\tilde{D}(p+U) = p'$. $\operatorname{null}\tilde{D} = (\operatorname{null} D)/U = U/U = \{0\}$, so $\tilde{D}$ is injective. $\operatorname{range}\tilde{D} = \operatorname{range} D = \mathcal{P}_2(\mathbb{R})$. So $\mathcal{P}_3(\mathbb{R})/\mathcal{P}_0(\mathbb{R}) \cong \mathcal{P}_2(\mathbb{R})$, with inverse $q \mapsto \int_0^x q + U$.

d) $p(0) = p'(0) = 0$ implies $p(0) = 0$, so $W \subseteq U$. $W = \operatorname{span}(x^2, x^3)$ and $U = \operatorname{span}(x, x^2, x^3)$, so $\dim V/W = 4-2 = 2$ and $\dim U/W = 3-2 = 1$. The map $p \mapsto p(0)$ is linear and surjective onto $\mathbb{R}$ with null space U, so $p + U \mapsto p(0)$ is an isomorphism $V/U \to \mathbb{R}$.

2. Linear functionals and dual maps.

a) $\operatorname{range}\varphi$ is a subspace of the 1-dimensional space $\mathbb{F}$, so it is $\{0\}$ or $\mathbb{F}$. If $\varphi \neq 0$, $\dim \operatorname{range}\varphi = 1$ and rank–nullity gives $\dim \operatorname{null}\varphi = \dim V - 1$.

b) If $\varphi = 0$, then $\operatorname{null}\psi = V$, so $\psi = 0 = 1 \cdot \varphi$. Otherwise pick v with $\varphi(v) = 1$ and set $c = \psi(v)$. Then $\psi - c\varphi$ is 0 on $\operatorname{null}\varphi = \operatorname{null}\psi$ and at v. Every $x \in V$ is $\big(x - \varphi(x)v\big) + \varphi(x)v$ with the first term in $\operatorname{null}\varphi$, so $\psi - c\varphi = 0$.

c) Let $A = \mathcal{M}(T)$, so $Tv_k = \sum_j A_{jk} w_j$. Then $(T'\psi_j)(v_k) = \psi_j(Tv_k) = A_{jk}$. By the coordinate formula $\chi = \sum_k \chi(v_k)\varphi_k$, $T'\psi_j = \sum_k A_{jk}\varphi_k$. So column j of $\mathcal{M}(T')$ is $(A_{j1}, \dots, A_{jn})$, which is row j of A: $\mathcal{M}(T') = A^T$.

d) $D'(\varphi_j)(x^k) = \varphi_j(kx^{k-1})$, which is k if $j = k-1$ and 0 otherwise. So $D'(\varphi_0) = \varphi_1$, $D'(\varphi_1) = 2\varphi_2$, $D'(\varphi_2) = 0$. (Check: $\mathcal{M}(D) = \left(\begin{smallmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{smallmatrix}\right)$, and its transpose has columns $(0,1,0)$, $(0,0,2)$, 0.)

3. Commuting operators and similarity.

a) If $Tv = \lambda v$, then $T(Sv) = S(Tv) = \lambda Sv$, so $Sv \in E(\lambda, T)$.

b) Let $\lambda_1, \dots, \lambda_n$ be the eigenvalues with eigenvectors $v_1, \dots, v_n$; these are independent, hence a basis, and $\sum \dim E(\lambda_i, T) \leq n$ with each term ≥ 1 forces $\dim E(\lambda_i, T) = 1$. By (a), $Sv_i \in E(\lambda_i, T) = \operatorname{span}(v_i)$, so $Sv_i = \mu_i v_i$: v_i is an eigenvector of S. (Any eigenvector of T is a multiple of some v_i.) The basis $v_1, \dots, v_n$ is an eigenbasis for S.

c) $v \in E(\lambda, R) \iff S^{-1}TSv = \lambda v \iff T(Sv) = \lambda Sv \iff Sv \in E(\lambda, T) \iff v \in S^{-1}\big(E(\lambda, T)\big)$. S^{-1} is injective, so $\dim E(\lambda, R) = \dim E(\lambda, T)$; in particular one is nonzero exactly when the other is.

d) By (b), e_1, e_2, e_3 are eigenvectors of S, so S is diagonal; conversely diagonal matrices commute with T. Directly: $(ST)_{ij} = s_{ij}t_j$ and $(TS)_{ij} = t_i s_{ij}$, so $s_{ij}(t_j - t_i) = 0$ and $s_{ij} = 0$ for $i \neq j$. Answer: all diagonal matrices.

4. The degree of the minimal polynomial.

a) $m_T(T|_U)u = m_T(T)u = 0$ for $u \in U$, so $m_T(T|_U) = 0$ and $m_{T|_U}$ divides m_T. Since $p(T/U)(v+U) = p(T)v + U$, $m_T(T/U)(v+U) = 0 + U$, so $m_T(T/U) = 0$ and $m_{T/U}$ divides m_T.

b) The $n+1$ vectors $v, Tv, \dots, T^n v$ are dependent, which gives a nonzero q with $\deg q \leq n$ and $q(T)v = 0$. If $u \in U$, then $q(T)Tu = Tq(T)u = 0$, so $Tu \in U$. If $q(T) = 0$, then m_T divides q, so $\deg m_T \leq \deg q \leq n$.

c) If $q(T) \neq 0$, then $U \neq V$, and $v \in U$ gives $U \neq \{0\}$; so $\dim U < n$ and $\dim V/U < n$. By induction $\deg r \leq \dim U$ and $\deg p \leq \dim V/U$. From $p(T/U) = 0$: $p(T)x + U = 0 + U$, so $p(T)x \in U$ for every $x \in V$. Then $r(T)p(T)x = r(T|_U)\big(p(T)x\big) = 0$. So $(rp)(T) = 0$, m_T divides rp, and $\deg m_T \leq \deg r + \deg p \leq \dim U + \dim V/U = n$.

d) $Ae_1 = e_2$, $Ae_2 = e_3$, so $e_1, Ae_1, A^2e_1 = e_1, e_2, e_3$ is a basis. $A^3e_1 = Ae_3 = (6, -11, 6) = 6e_1 - 11Ae_1 + 6A^2e_1$, so $q = x^3 - 6x^2 + 11x - 6$ has $q(A)e_1 = 0$, hence $q(A)A^k e_1 = A^k q(A)e_1 = 0$: $q(A) = 0$ on the basis. A nonzero polynomial of degree ≤ 2 cannot kill e_1 (that would be a dependence among e_1, e_2, e_3), so $m_A = q = (x-1)(x-2)(x-3)$. Distinct roots: A is diagonalizable, with eigenvalues $1, 2, 3$.